

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

**(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)**

Department of Computer Engineering



Project Report on

StockSense AI: Your Personal Investment Strategist

In partial fulfillment of the Fourth Year (Semester–VII), Bachelor of Engineering
(B.E.) Degree in Computer Engineering at the University of Mumbai

Academic Year 2025-26

Ms. Pallavi Saindane

Submitted by

Puneet Singh Kewalramani
(D17A - 36)

Chetan Narang
(D17C - 45)

Vivek Menghani
(D17B - 71)

Manit Khira
(D17B - 22)

(2025-26)

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)

Department of Computer Engineering



CERTIFICATE of Approval

This is to certify that **Puneet Singh Kewalramani (D17A - 36), Chetan Narang (D17C - 45), Mani Khira (D17B - 22), Vivek Menghani (D17B - 71)** of Fourth Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on **StockSense AI: Your Personal Investment Strategist** as a part of the coursework of PROJECT-I for Semester-VII under the guidance of **Ms. Pallavi Saindane**, in the year 2025-2026.

Date

Internal Examiner

External Examiner

Project Mentor

Head of the Department
Dr. Mrs. Nupur Giri

Principal
Dr. J. M. Nair

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea / data / fact / source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

(Signature)

(Name of student and Roll No.)

(Signature)

(Name of student and Roll No.)

(Signature)

(Name of student and Roll No.)

(Signature)

(Name of student and Roll No.)

Date:

ACKNOWLEDGEMENT

We are thankful to our college Vivekanand Education Society's Institute of Technology for considering our project and extending help at all stages needed during our work of collecting information regarding the project.

It gives us immense pleasure to express our deep and sincere gratitude to Assistant Professor **Prof. Ms. Pallavi Saindane** (Project Guide) for her kind help and valuable advice during the development of project synopsis and for her guidance and suggestions.

We are deeply indebted to Head of the Computer Department **Dr.(Mrs.) Nupur Giri** and our Principal **Dr. (Mrs.) J.M. Nair** , for giving us this valuable opportunity to do this project.

We express our hearty thanks to them for their assistance without which it would have been difficult in finishing this project synopsis and project review successfully.

We convey our deep sense of gratitude to all teaching and non-teaching staff for their constant encouragement, support and selfless help throughout the project work. It is a great pleasure to acknowledge the help and suggestion, which we received from the Department of Computer Engineering.

We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

Computer Engineering Department

COURSE OUTCOMES FOR B.E PROJECT

Learners will be to:-

Course Outcome	Description of the Course Outcome
CO 1	Do literature survey/industrial visit and identify the problem of the selected project topic.
CO2	Apply basic engineering fundamental in the domain of practical applications FOR problem identification, formulation and solution
CO 3	Attempt & Design a problem solution in a right approach to complex problems
CO 4	Cultivate the habit of working in a team
CO 5	Correlate the theoretical and experimental/simulations results and draw the proper inferences
CO 6	Demonstrate the knowledge, skills and attitudes of a professional engineer & Prepare report as per the standard guidelines.

ABSTRACT

Navigating the complexities of the stock market presents a significant challenge for investors, particularly when selecting an appropriate investment strategy for different market segments. The effectiveness of a given strategy can vary dramatically between large-cap, mid-cap, and small-cap stocks, yet investors often lack the tools to validate these approaches. This leads to uncertainty and decisions based on generic advice rather than data-driven evidence tailored to specific market conditions.

To address this challenge, we propose StockSense AI, an intelligent stock market advisory platform designed to provide personalized, strategy-specific guidance. The system analyzes historical market data to evaluate the performance of various investment strategies across different market-cap segments. The core of the platform is a powerful backtesting engine that simulates historical strategy performance, allowing users to see how a strategy would have performed on a specific stock. Based on this analysis, the platform generates data-backed insights, helping users understand a strategy's historical viability and risk-return profile.

Built using Python and the Streamlit framework, the platform features an interactive dashboard with dynamic visualizations. It currently implements foundational strategies like Moving Average Crossover, RSI, and Bollinger Bands. By providing clear, actionable, and personalized historical evidence, StockSense AI empowers users to make more informed and confident investment decisions, effectively bridging the gap between theoretical strategies and practical application.

Table of Contents

Certificate.....	i
Declaration.....	ii
Acknowledgment.....	iii
Abstract.....	iv
Table of Contents.....	v
Chapter 1: Introduction.....	1
1.1 Introduction to the Project.....	1
1.2 Motivation for the Project.....	2
1.3 Problem Definition.....	3
1.4 Relevance of the Project.....	4
1.5 Methodology Used.....	5
Chapter 2: Literature Survey.....	6
2.1 Research Papers Referred (Abstracts & Inferences).....	6
2.2 Books / Journals / Articles Referred.....	9
2.3 Interactions with Domain Experts.....	10
2.4 Patent Search.....	11
2.5 Existing Systems.....	13
2.6 Lacuna in the Existing Systems.....	14
2.7 Comparison of Existing Systems and Proposed Area of Work.....	15
2.8 Focus Area.....	16
Chapter 3: Requirements for the Proposed System.....	17

3.1 Functional Requirements.....	17
3.2 Non-Functional Requirements.....	17
3.3 Constraints.....	18
3.4 Hardware & Software Requirements.....	18
3.5 Techniques Utilized.....	18
3.6 Tools Utilized.....	19
3.7 Project Proposal.....	19
 Chapter 4: Proposed Design.....	 20
4.1 Block Diagram of the Proposed System.....	20
4.2 Modular Diagram.....	21
4.3 Detailed Design.....	22
4.3.1 DFD – Level 0 (Context Diagram).....	22
4.3.2 DFD – Level 1.....	23
4.3.3 Flow Diagram.....	24
4.3.4 Screenshot of Implementation.....	25
4.4 Algorithms Utilized.....	31
 Chapter 5: Proposed Results and Discussions.....	 32
5.1 Determination of Efficiency.....	32
5.2 Determination of Accuracy.....	32
5.3 Reports on Sensitivity Analysis.....	33
5.4 Graphs: Accuracy vs Time.....	34
 Chapter 6: Plan of Action for the Next Semester.....	 35

6.1 Work Done Till Date.....	35
6.2 Plan of Action for Project II.....	35
 Chapter 7: Conclusion.....	 36
 Chapter 8: References.....	 37
 Chapter 9: Appendix.....	 39
9.1 List of Figures.....	39
9.2 List of Tables.....	40

Chapter 1: Introduction

1.1 Introduction to the Project

The Indian stock market, with over 8 crore (80 million) registered investors as of 2024, represents a massive yet often overwhelmed participant base. A significant majority of these investors are retail participants who navigate the complexities of equity trading without the sophisticated tools available to institutional players. The core challenge lies not in a lack of strategies—concepts like Moving Average crossovers or RSI-based trading are widely known—but in the critical context of where and when these strategies are effective. A momentum-based approach might yield substantial returns in a volatile small-cap stock but could lead to significant losses if applied to a stable, large-cap blue-chip, and vice-versa.

StockSense AI is conceived as an intelligent, AI-powered advisory platform designed to bridge this critical gap between theoretical knowledge and practical, profitable application. The project moves beyond being a mere data visualization tool by establishing itself as an interactive strategy validation ecosystem. Its primary function is to empower investors by providing a powerful backtesting engine that simulates the historical performance of various trading strategies across different market segments, including Large-Cap, Mid-Cap, and Small-Cap stocks.

Built on a robust technology stack utilizing Python for computational logic and Streamlit for an accessible web-based dashboard, the platform demystifies strategy performance. It allows users to not only view historical price data through interactive charts but also to apply technical indicators like Simple Moving Averages (SMA), Relative Strength Index (RSI), and Bollinger Bands, and then simulate the exact trades and portfolio outcomes these strategies would have generated in the past.

The vision for StockSense AI extends beyond retrospective analysis. The architectural foundation is designed to evolve into a personalized recommendation engine. In future iterations, the system aims to analyze a user's historical interactions and risk appetite to suggest optimal strategies and specific stocks, transforming from a validation tool into a proactive, intelligent advisory partner.

1.2 Motivation for the Project

The motivation for StockSense AI stems from a clear and significant disparity in the financial technology landscape. Popular trading platforms offered by major brokers in India primarily focus on the execution of trades—providing real-time data, charting tools, and order placement facilities. However, they offer little to no guidance on strategy formulation or validation. This creates a dangerous environment where beginners, armed with knowledge but no practical validation means, often make decisions based on hearsay, social media tips, or emotional reactions, leading to substantial capital erosion.

Furthermore, the existing tools that do offer backtesting are often complex, expensive, and targeted at professional quants or fund managers, putting them out of reach for the average retail investor. This creates an "information asymmetry" where institutional investors leverage high-end analytics while retail investors are left with generic, one-size-fits-all advice.

StockSense AI is motivated by the need to democratize these advanced analytical capabilities. The project is driven by the following core principles:

Bridging Theory and Practice: To transform abstract trading concepts from textbooks into tangible, testable simulations using real-world market data.

Promoting Data-Driven Decisions: To shift the investor's paradigm from relying on gut feeling and generic advice to making decisions grounded in historical evidence and quantitative analysis.

Enhancing Financial Literacy: To serve as an educational platform where users can experiment, learn from historical outcomes (both profitable and loss-making), and develop a deeper understanding of market dynamics and risk management.

By providing these capabilities through an intuitive and user-friendly dashboard, StockSense AI aims to level the playing field, fostering a more informed and confident investing community.

1.3 Problem Definition

The current ecosystem of stock market advisory and analysis tools is plagued by several critical shortcomings that hinder an investor's ability to make informed decisions. The problem is multi-faceted, impacting both novice and experienced traders.

Current Challenges in Stock Strategy Analysis:

Generic, Non-Segmented Recommendations: Existing advisory systems and financial news portals often provide blanket recommendations (e.g., "buy," "sell," "hold") that fail to account for the fundamental differences in risk, volatility, and price behavior across market capitalization segments (Large-Cap, Mid-Cap, Small-Cap). A strategy's performance is highly context-dependent, a nuance completely missed by current systems.

Lack of Accessible Backtesting Tools: Investors have no straightforward means to empirically validate a trading strategy against historical data. They are forced to either rely on theoretical understanding or use complex programming libraries which require significant technical expertise, creating a high barrier to entry.

Absence of Personalized Insights: Current platforms do not tailor their analysis to the individual user. There is no mechanism to consider a user's unique risk tolerance, investment horizon, or historical preferences, leading to a disconnect between the advice given and the investor's personal financial goals.

Over-reliance on Static Charts and Theoretical Knowledge: While charts displaying technical indicators are common, they are typically static and non-interactive. Users cannot tweak parameters or see the direct, simulated financial impact of a strategy, leaving a gap between observation and actionable insight.

Problem Statement:

To design and develop an intelligent, interactive, and AI-powered stock market advisory system that enables investors to select, rigorously backtest, and visually analyze the performance of various trading strategies across different market-cap segments. The system will replace generic

advice with data-driven, personalized insights, allowing users to validate strategy viability through historical simulation and make more confident, informed investment decisions.

1.4 Relevance of the Project

StockSense AI is profoundly relevant in the contemporary context of India's rapidly growing retail participation in equity markets. Its impact extends across multiple dimensions:

Societal Impact (Financial Inclusion & Literacy): The platform acts as a force multiplier for financial literacy. By providing a hands-on, experimental environment, it educates users on the practical application of technical analysis, promoting a disciplined and evidence-based approach to investing over speculative gambling. This aligns with national efforts to enhance financial awareness.

Technological Impact (Democratizing AI & Analytics): The project uses technologies like Python-based data analytics and interactive frameworks to build a tool that is typically available only to professionals. It demonstrates how technology can be harnessed to create inclusive solutions that empower the average individual.

Economic Impact (Informed Decision Making): By enabling better, data-backed investment decisions, the platform has the potential to improve individual portfolio returns and reduce capital loss. A more informed investor base contributes to a more stable and efficient market overall.

Foundational for Future Innovation: The platform is not a static product but a foundation. Its modular architecture allows for the seamless integration of advanced AI models for forecasting and personalization in the future, positioning it at the forefront of the fintech innovation curve.

1.5 Methodology Used

The development of StockSense AI follows a structured, multi-stage methodology to ensure a robust and user-centric platform. The development of StockSense AI follows a structured, multi-stage methodology to ensure a robust, reliable, and user-centric platform. This comprehensive approach is designed to systematically address all aspects of the project, from initial conceptualization and data acquisition to model training, deployment, and continuous improvement. The methodology emphasizes a cyclical and iterative process, allowing for regular feedback integration and adjustments to optimize performance and user experience.

Data Acquisition & Processing:

A curated list of NSE-listed stocks, categorized by market cap, forms our analytical universe. Historical OHLCV data is dynamically fetched via the yfinance API. The raw data undergoes cleaning and processing, with key technical indicators (SMA, RSI, Bollinger Bands) calculated for strategy implementation.

Strategy Implementation & Backtesting:

Core trading strategies are algorithmically encoded using Python's pandas library. The backtesting engine simulates trades based on generated signals, allowing users to customize parameters like MA windows and RSI levels. The system tracks all trades and calculates performance metrics including Total Return, Final Portfolio Value, and Win Rate.

Visualization & Dashboard:

Interactive charts powered by Plotly display price trends, technical indicators, and buy/sell signals. The entire system is integrated into a multi-page web application using Streamlit, providing a seamless user experience for strategy testing and analysis.

Evaluation & Future Scope:

The platform enables comparative strategy analysis across different market segments, while establishing a foundation for future AI-powered personalized recommendations based on user behavior and risk profiles.

Chapter 2: Literature Survey

2.1 Research Papers Referred (Abstracts & Inferences)

[1] Z. Chen and J. Lee, “Enhancing Financial Trading Strategies using Transformer-based Deep Learning Models,” IEEE SSCI, 2024.

Abstract:

The paper explores how transformer-based models can improve financial trading by capturing long-term dependencies in time-series data. It demonstrates how attention mechanisms help detect patterns that traditional models may miss, resulting in improved trend prediction and strategy performance.

Inference:

This highlights the growing importance of sequence-aware models. Although our system currently uses rule-based technical strategies, the paper guides future expansion toward transformer-based personalized recommendations.

[2] Y. Wang, Q. Liu, and J. Liang, “A Multi-Objective Deep Reinforcement Learning Framework for Algorithmic Trading,” IEEE TNNLS, 2024.

Abstract:

The authors propose a reinforcement learning model that optimizes trading decisions based on multiple objectives such as profit, risk, and transaction cost. The framework adapts to changing market conditions and learns effective policies through repeated interaction with market data.

Inference:

This supports the idea that different market segments (Large/Mid/Small-Cap) require different strategies. It reinforces our goal of segment-specific backtesting and later integrating adaptive recommendation modules.

[3] H. Liu and J. Wu, “A Hybrid CNN-GRU Model with Attention Mechanism for Stock Market Trend Prediction,” IEEE Access, 2024.

Abstract:

This study combines convolutional neural networks and GRUs with an attention layer to identify

short-term and long-term market patterns. The hybrid model outperforms traditional machine learning approaches in directional trend prediction.

Inference:

The paper demonstrates how hybrid deep learning models enrich stock movement analysis. For our system, it reinforces the importance of accurate data preprocessing and time-series handling, even for rule-based strategies.

[4] L. Yang, Z. Li, and D. Wu, “Sector-Specific Stock Trend Prediction Using Graph Neural Networks,” IEEE Big Data, 2023.

Abstract:

The research introduces a graph neural network (GNN) model for sector-wise stock prediction by treating companies as interconnected entities. It captures cross-sector relationships to improve forecast accuracy.

Inference:

This directly supports our system’s feature of categorizing stocks by market segments, highlighting that segmentation improves analytical relevance and strategy interpretation.

[5] R. Kumar and S. Roy, “A Goal-Based Robo-Advising System using Deep Learning for Personalized Asset Allocation,” IEEE Access, 2024.

Abstract:

The paper presents a robo-advisor that tailors investment portfolios based on user goals using deep learning models. It factors age, income, and risk tolerance to generate allocations dynamically.

Inference:

This aligns with our future direction of building a subtle recommendation module inside StockSense AI, offering user-personalized insights instead of generic strategy results.

[6] A. Gupta and P. Singh, “An Event-Driven Backtesting Framework for High-Frequency Trading Strategies,” IEEE CASE, 2023.

Abstract:

The authors design an event-driven system for evaluating high-frequency strategies by simulating trades based on market micro-events. Their backtester prioritizes speed and accuracy

at millisecond resolution.

Inference:

It validates the importance of an efficient backtesting engine. Although our system operates at daily/weekly resolution, the modular structure aligns with event-driven best practices.

[7] F. Ahmad, “Machine Learning Models for Client Risk Profiling in Digital Wealth Management,” IEEE TEM, 2023.

Abstract:

This work focuses on using ML models to classify clients by risk tolerance using behavioral and financial data. It demonstrates how automated tools can improve advisory accuracy.

Inference:

Supports our long-term aim of personalized recommendations by showing how user profiling enhances strategy suitability in financial systems.

[8] J. Xu, C. Zhou, and Y. Luo, “Meta Reinforcement Learning for Trading in Non-Stationary Markets,” IASc, 2024.

Abstract:

The paper applies meta-RL to handle rapidly changing market conditions by enabling models to learn how to adapt their learning strategies.

Inference:

Highlights the reality that strategies perform differently across market regimes—a core problem our backtesting engine helps users understand.

[9] R. Wang, S. Li, and H. Zhang, “Deep RL for Stock Trading With Multi-Modal Features,” Expert Systems with Applications, 2024.

Abstract:

The authors integrate technical indicators, news sentiment, and price data into a multi-modal DRL model, enhancing trading performance in volatile markets.

Inference:

This reinforces the potential for expanding our platform later to include multi-modal data (news, sentiment) to enhance strategy accuracy and recommendations.

2.2 Books / Journals / Articles Referred

[10] J. J. Murphy, *Technical Analysis of the Financial Markets: A Comprehensive Guide to Trading Methods and Applications*, New York Institute of Finance, 1999.

This book is considered the foundational text for technical analysis. It explains chart patterns, indicators, market psychology, and trend behavior in detail. Concepts such as moving averages, RSI, and Bollinger Bands used in our system are based directly on Murphy's formulations. The book helped establish the theoretical basis for the strategies implemented in StockSense AI.

[11] S. Nison, *Japanese Candlestick Charting Techniques*, 2nd ed., New York Institute of Finance, 2001.

Nison's work popularized candlestick charting in Western markets. It provides extensive explanations of candlestick patterns, trend reversals, and visual cues in price movement. This book informed the design of our system's candlestick visualizations and helped in selecting chart types that communicate market behavior clearly to users.

[12] M. Douglas, *Trading in the Zone: Master the Market with Confidence, Discipline, and a Winning Attitude*, Prentice Hall Press, 2000.

Douglas focuses on trader psychology, risk perception, and disciplined decision-making. While not directly technical, the book emphasized the need for clarity, transparency, and learning-oriented tools. This influenced our platform's educational design approach—helping beginners understand strategy behavior rather than blindly following indicators.

Web Platforms / Technical Articles

[13] Yahoo Finance API Documentation (yfinance), Yahoo Inc., 2024. [Online]. Available: <https://pypi.org/project/yfinance/>

Provided reference for retrieving OHLCV stock data, including API usage, parameter settings, and data structures. This documentation ensured accurate data fetching and preprocessing in the platform.

[14] Streamlit Documentation, Streamlit Inc., 2024. [Online]. Available: <https://docs.streamlit.io/>

Used to understand the framework for building interactive dashboards, handling user input, and structuring UI components. It helped design a user-friendly interface for strategy selection and backtesting.

[15] Plotly Python Graphing Library, Plotly Technologies Inc., 2024. [Online]. Available: <https://plotly.com/python/>

Explains charting functions, interactive visual elements, and customization options. It guided the creation of line charts, candlestick charts, signal overlays, and equity curves within StockSense AI.

2.3 Interactions with Domain Experts

During the development of the project, exposure to domain knowledge was obtained through faculty guidance and expert-led online sessions. These interactions—though not formal one-to-one consultations—provided useful context for understanding trading strategies, market behavior, and practical considerations in financial analytics.

1. Faculty Discussions:

Guidance from project mentors helped refine the scope of StockSense AI, ensuring that the system remained technically feasible while maintaining academic depth. Inputs from faculty were instrumental in structuring the modular design and identifying suitable strategies for implementation.

2. Webinar by Hardik Raja, Co-founder of Skill Nation:

The project team attended an online webinar conducted by Hardik Raja, a practitioner in algorithmic trading and quantitative finance. The session introduced fundamental concepts such as strategy validation, market regime differences, and the importance of backtesting. Although not all aspects of the session were followed in depth, the webinar reinforced the need for transparent, data-driven tools—supporting our aim to create an educational backtesting platform.

3. Online Expert Resources and Industry Material:

Insights were drawn from publicly available talks, interviews, and blogs by well-known

financial analysts and quantitative traders. Content from platforms like Zerodha Varsity, TradingView community experts, and AlgoTrading India provided additional clarity on technical indicators, risk management, and common pitfalls in retail trading.

4. **Technical Documentation from Industry Tools:**

Expert-authored documentation provided by Yahoo Finance (yfinance), Streamlit, and Plotly offered practical guidance for implementing data pipelines, visual dashboards, and indicator overlays. These resources served as indirect expert input for designing a robust and user-friendly analytical system.

While direct in-person consultations with industry professionals were not conducted, the combination of faculty guidance, expert webinars, and authoritative online resources contributed valuable insights and helped shape the direction of the proposed system.

2.4 Patent Search

As part of the literature review, a patent search was conducted using the official repositories—Espacenet (European Patent Office) and the United States Patent and Trademark Office (USPTO)—to identify innovations related to trading strategy automation, portfolio optimization, and backtesting frameworks. Four patents were shortlisted due to their relevance to algorithmic trading systems, data-driven decision support, and financial analytics. The key insights and inferences drawn from each are summarized below.

1. **Derivative Trading Strategy Backtesting Machine (US20100153300A1)**

This patent proposes a comprehensive backtesting system that simulates derivative trading strategies using historical market data [13]. The invention focuses on enabling users to compare multiple strategies, analyze profitability, and generate performance metrics across varying market conditions.

Inference: The patent highlights the importance of a modular simulation engine and structured data pipeline. This reinforces the design choice in our system to incorporate independent modules for data fetching, signal generation, and backtesting, ensuring flexibility and scalability.

2. **Instantly Back-Testing Trading Strategies in an Options Portfolio (US20140379551A1)**

This invention introduces a system for real-time or near-instant backtesting of option-based strategies [14]. It emphasizes computational efficiency and rapid evaluation using optimized data structures and parallel processing.

Inference: The patent underscores the need for fast, user-responsive backtesting mechanisms. Although our system targets beginners and uses simpler instruments (equities instead of options), the concept of rapid simulation validates our goal of achieving efficient backtests within a few seconds.

3. **System for Computing Optimized Trading Instructions Using an Ensemble of Prediction Models (US20210082046A1)**

This patent details a framework that uses multiple predictive models to generate optimized trading instructions [15]. It integrates machine learning ensembles, risk modeling, and performance tracking to enhance decision-making accuracy.

Inference: While our system currently implements rule-based technical strategies, this patent demonstrates a path toward machine-learning-driven personalization. This supports the longer-term vision of evolving StockSense AI into a recommendation-oriented advisory platform.

4. **Automated Financial Portfolio Recommendation and Risk Assessment Platform (US20190245613A1)**

This system provides automated portfolio recommendations using user profiles, market analytics, and risk metrics [16]. It incorporates data ingestion pipelines and automated scoring mechanisms to generate suitable asset allocations.

Inference: This invention reinforces the role of user-centered personalization in modern financial systems. It aligns with our project's extended goal of eventually offering tailored recommendations based on user preferences, risk tolerance, and backtested behaviors.

These patents collectively validate the relevance of modular design, fast computation, educational backtesting, and personalization as meaningful directions for financial analytics systems. They also provide useful architectural inspiration while ensuring that the proposed system remains distinct, academically original, and beginner-focused.

2.5 Existing Systems

The current ecosystem of stock market analytics and strategy-testing tools is dominated by several widely used platforms, each catering to different user groups and offering specific functionalities:

Zerodha [16]:

A leading discount brokerage in India that focuses primarily on seamless trade execution. While it offers basic charting through its Kite platform, its backtesting capabilities are minimal and mostly restricted to simple indicators. It is built for execution rather than education or strategy experimentation.

TradingView [17]:

A globally popular charting and analysis platform known for its advanced indicators, Pine Script-based strategy development, and social community features. However, TradingView largely serves intermediate to advanced traders, and its backtesting tools require scripting knowledge, making it less suitable for beginners seeking guided learning.

Streak [18]:

A no-code strategy-building and backtesting platform that integrates with Zerodha and other brokers. Although beginner-friendly compared to TradingView, it still assumes a foundational understanding of indicators and trading logic. Its backtesting is strong but not personalised or educational.

Dhan:

An emerging trading platform in India that provides API access and some analytical tools. However, its ecosystem is still evolving, and it lacks comprehensive learning-oriented backtesting modules for beginners.

Overall, these systems either prioritize **execution (Zerodha, Dhan)** or **technical charting for experienced users (TradingView, Streak)**. There remains a clear gap for platforms that simplify concepts and encourage experimentation without requiring deep technical knowledge.

2.6 Lacuna in the Existing Systems

A critical analysis of current platforms reveals several important gaps that limit accessibility for new traders:

Beginner-Unfriendly Interfaces:

Most systems are designed for users already familiar with financial terminology, indicators, and chart patterns. This makes the learning curve steep and discourages new investors.

Lack of Educational Orientation:

Platforms are optimized for placing trades or analyzing markets rather than teaching strategy logic. Users can follow signals, but rarely understand *why* strategies perform a certain way.

Generic Strategy Recommendations:

Existing systems do not adjust their suggestions based on market categories like large-cap, mid-cap, or small-cap stocks. This results in broad, non-contextual advice that may not apply uniformly across segments.

Complex or Inaccessible Backtesting:

Backtesting options either require programming skills (e.g., Pine Script in TradingView) or involve multiple technical configurations that beginners may find overwhelming.

No Personalized Learning or Feedback:

Current tools do not adapt to user behavior, past strategy choices, or learning patterns. Users receive the same fixed set of tools regardless of their experience level.

These gaps highlight the need for a platform that prioritizes accessibility, education, and guided experimentation, forming the basis for StockSense AI.

2.7 Comparison of Existing Systems and Proposed Area of Work

Feature	Zerodha	TradingView	Streak	StockSense AI (Proposed)
Target Audience	All traders	Experienced traders	Tech-savvy traders	Beginner to Intermediate
Backtesting	Limited	Community scripts	Advanced	Educational & Interactive
Personalization	None	None	Limited	AI-Powered Recommendations
Market Segmentation	No	No	No	Yes (Large/Mid/Small-Cap)
Educational Focus	Low	Medium	Low	High
Code Requirements	None	Pine Script	No-code	No-code Interface

Table 2.7

2.8 Focus Area

The proposed system aims to address the gaps identified above through a focused set of objectives:

Democratizing Analysis:

StockSense AI simplifies strategy validation, enabling retail investors to test trading ideas without needing programming or complex software skills. This lowers the barrier for financial literacy.

Segment-Specific Insights:

Instead of treating all stocks the same, the system emphasizes comparisons across market capitalizations—Large-Cap, Mid-Cap, and Small-Cap. This helps users understand how strategies behave differently across market segments.

Interactive Learning Environment:

By combining visual explanations, clean dashboards, and dynamic backtesting, the platform encourages hands-on exploration. Users can tweak parameters and immediately observe how strategy performance changes.

Foundation for Personalization:

Although the initial version focuses on educational backtesting, the architecture is designed to support future enhancements such as AI-based strategy recommendations, user profiling, and adaptive learning modules.

This focused approach positions StockSense AI as an accessible, educational, and incremental system that can grow into a personalized trading assistant in later stages.

Chapter 3: Requirements for the Proposed System

3.1 Functional Requirements

User Registration & Authentication - Essential for securing personalized user data and maintaining individual investment profiles and strategy histories.

Stock Data Management - Critical for providing organized access to categorized NSE stocks, enabling targeted analysis across different market segments and sectors

Strategy Implementation - Fundamental to offering diverse technical analysis approaches (MA Crossover, RSI, Bollinger Bands) that form the core analytical capabilities.

Backtesting Engine - Core functionality that enables historical strategy validation with customizable parameters to assess performance under various market conditions.

Interactive Visualization - Vital for transforming complex financial data into intuitive charts and dashboards that facilitate informed investment decisions.

3.2 Non-Functional Requirements

Performance (10-second backtesting) - Crucial for maintaining user engagement through responsive analysis and near real-time results during strategy testing.

Usability (Intuitive interface) - Essential for accommodating both novice and experienced investors without requiring extensive training or technical expertise.

Reliability (99.5% uptime) - Necessary to ensure consistent platform availability for time-sensitive investment decisions and strategy analysis.

Security (Secure authentication) - Fundamental for protecting sensitive user data, investment strategies, and personal financial information.

3.3 Constraints

Python & Streamlit Framework - Selected for rapid prototyping capabilities, extensive financial libraries, and efficient web deployment.

yfinance API Dependency - Utilized for cost-effective access to reliable market data while accepting minor delays in exchange for zero licensing costs.

NSE Stocks Focus - Chosen to provide specialized analysis of Indian markets rather than diluted global coverage.

Open-source Technologies - Implemented to ensure cost-effectiveness and leverage community support for ongoing development.

3.4 Hardware & Software Requirements

Hardware Specifications - Standard computing resources sufficient for data processing and backtesting computations without requiring specialized infrastructure.

Python 3.9+ Environment - Provides essential libraries for financial analysis while maintaining machine learning readiness.

Modern Web Browsers - Ensure full compatibility with Streamlit's interactive components and Plotly visualization capabilities.

3.5 Techniques Utilized

Object-Oriented Programming - Enables modular, maintainable code architecture for strategy implementation and data management.

Algorithmic Trading Logic - Implements precise technical analysis rules to ensure accurate and reliable signal generation.

Backtesting Methodology - Provides rigorous historical validation framework for assessing trading strategy performance.

Time-Series Analysis - Handles sequential stock data efficiently to identify patterns and trends in market behavior.

3.6 Tools Utilized

Python Ecosystem - Offers comprehensive data manipulation libraries (Pandas, NumPy) essential for financial computations.

Streamlit Framework - Enables rapid development of interactive web applications using Python backend.

yfinance API - Provides free, reliable access to historical market data with comprehensive coverage.

Plotly Visualization - Delivers professional, interactive charting capabilities for financial data presentation.

Git/GitHub - Ensures robust version control and supports collaborative development processes.

3.7 Project Proposal

StockSense AI addresses the critical market gap in accessible strategy validation tools by providing retail investors with a comprehensive platform that transforms theoretical knowledge into practical, testable strategies. The system enables risk mitigation through pre-trade validation and promotes data-driven decisions by replacing emotional trading with evidence-based analysis. The architecture establishes a scalable foundation capable of integrating advanced AI features for personalized investment advisory, positioning the platform for future growth and enhanced functionality.

Chapter 4: Proposed Design

4.1 Block Diagram of the Proposed System

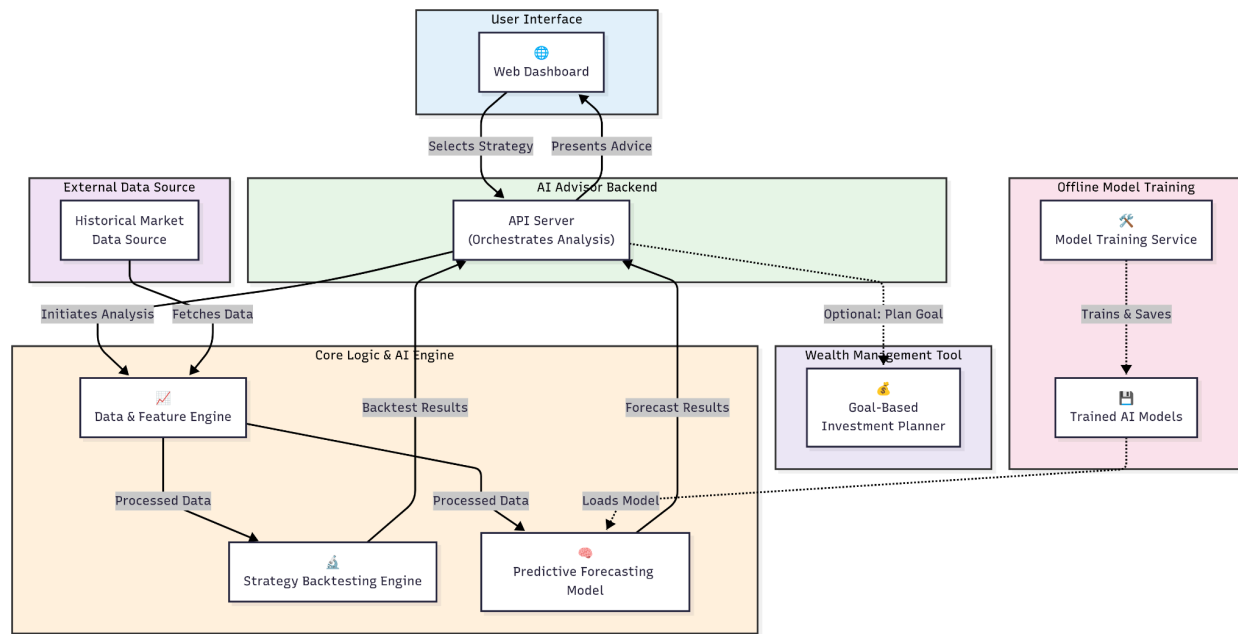


Fig. 4.1 Block Diagram

Explanation:

The block diagram illustrates the high-level architecture of StockSense AI as an intelligent advisory platform. The system begins with the User Interface (Web Dashboard) where investors interact with the platform. This interface connects to the AI Advisory Platform (Core Engine) that orchestrates all analysis. The core engine coordinates four main components: the Data & Feature Engine that fetches and processes market data from external sources; the Strategy Engine implementing technical analysis methods (MA Crossover, RSI, Bollinger Bands); the Backtesting Engine that simulates portfolio performance; and the Recommendation Engine that generates personalized strategy suggestions. Data flows sequentially through these components, with each module building upon the previous one's output, ultimately returning actionable insights to the user through the dashboard interface.

4.2 Modular Diagram

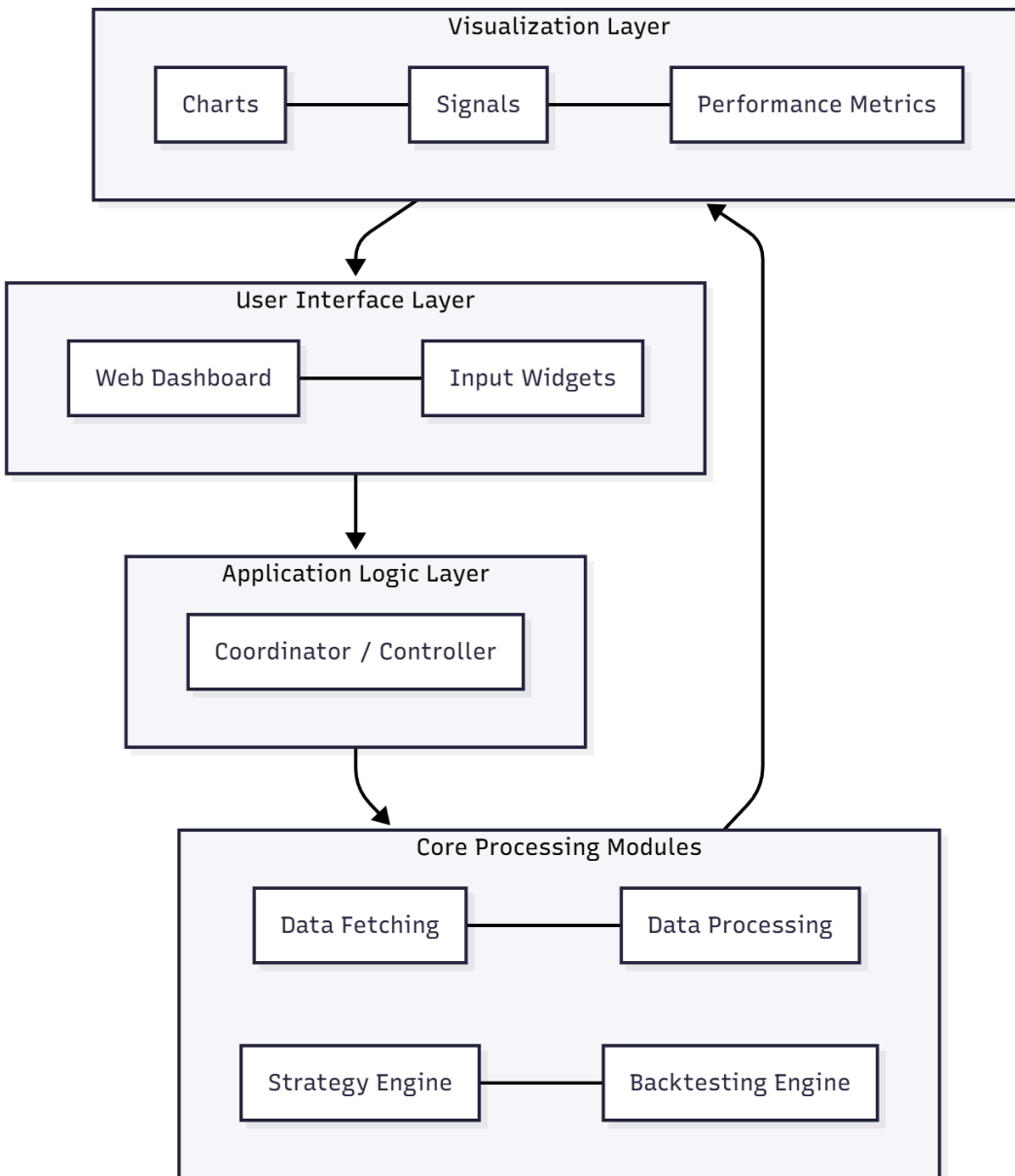


Fig. 4.2 Modular Diagram

The modular diagram illustrates the layered architecture of the StockSense AI system, showing how different components interact to enable stock analysis and backtesting. At the top, the Visualization Layer contains the modules responsible for presenting results to the user, including

charts, trading signals, and performance metrics. These outputs feed into and receive updates from the User Interface Layer, which provides an interactive web dashboard and input widgets through which users select stocks, strategies, and parameters. Below this, the Application Logic Layer acts as the central controller, coordinating data flow between the interface and the analytical backend. It forwards user inputs to the Core Processing Modules, which form the analytical engine of the system. This core layer includes modules for data fetching, data processing, strategy computation, and backtesting. Together, these modules retrieve market data, clean and transform it, apply the chosen trading strategy, generate signals, and simulate historical performance. The results are then passed upward to the Visualization Layer, completing the cycle. Overall, the diagram highlights a clean, modular workflow where each layer has a distinct responsibility, ensuring clarity, scalability, and seamless interaction between the user interface and backend analytics.

4.3 Detailed Design

4.3.1 DFD – Level 0 (Context Diagram)

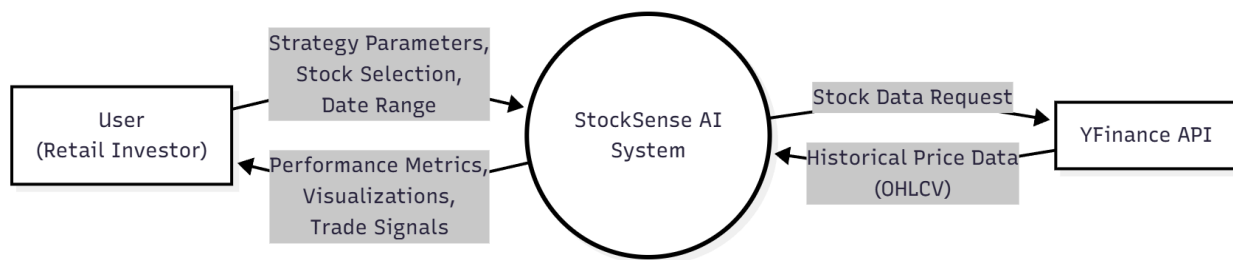


Fig. 4.3.1 (a) Data Flow Diagram - Level 0

The Level 0 Data Flow Diagram (context diagram) presents StockSense AI as a single, unified system interacting with external entities. The user provides inputs such as stock selection, strategy choice, and analysis parameters to the system, while the YFinance API supplies the required historical OHLCV price data. The system processes these inputs and returns outputs in the form of performance metrics, trade signals, and visualizations. This diagram captures the

highest-level view of data movement without showing internal components, helping establish the system boundaries and the main sources and destinations of information.

4.3.2 DFD – Level 1

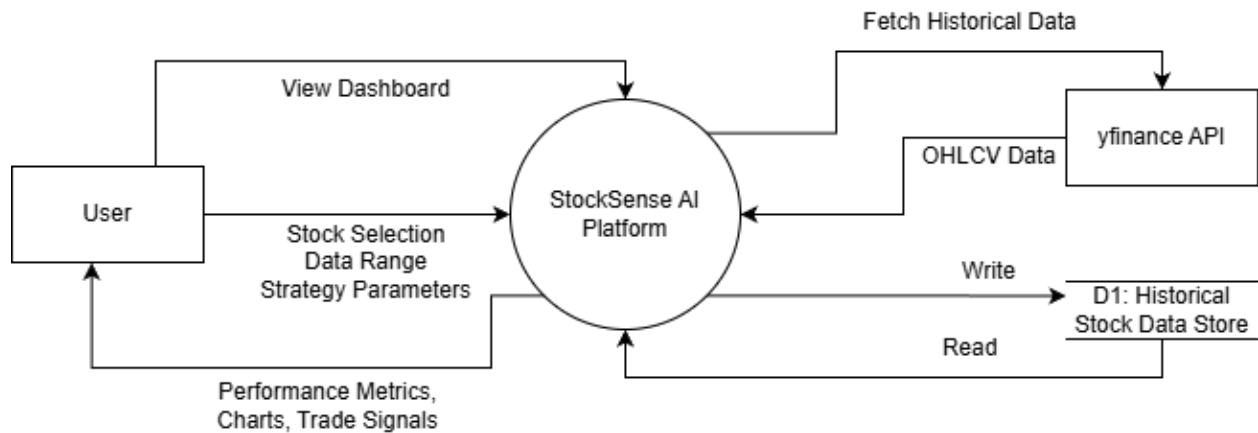


Fig. 4.3.2 (b) Data Flow Diagram - Level 1

The Level 1 Data Flow Diagram expands the internal workflow of the StockSense AI system into major functional processes. User inputs first enter the Input Handler, which forwards data requests to the Data Fetching module. Historical stock data obtained from the external API is sent to the Data Processing module for cleaning and transformation. The processed data flows into the Strategy and Backtesting Engine, where trading signals are generated and the strategy's historical performance is evaluated. Finally, the Visualization Generator converts these results into charts and metrics, which are returned to the user. This level of detail demonstrates how StockSense AI transforms raw market data into meaningful insights through sequential data flows.

4.3.3 Flow Diagram

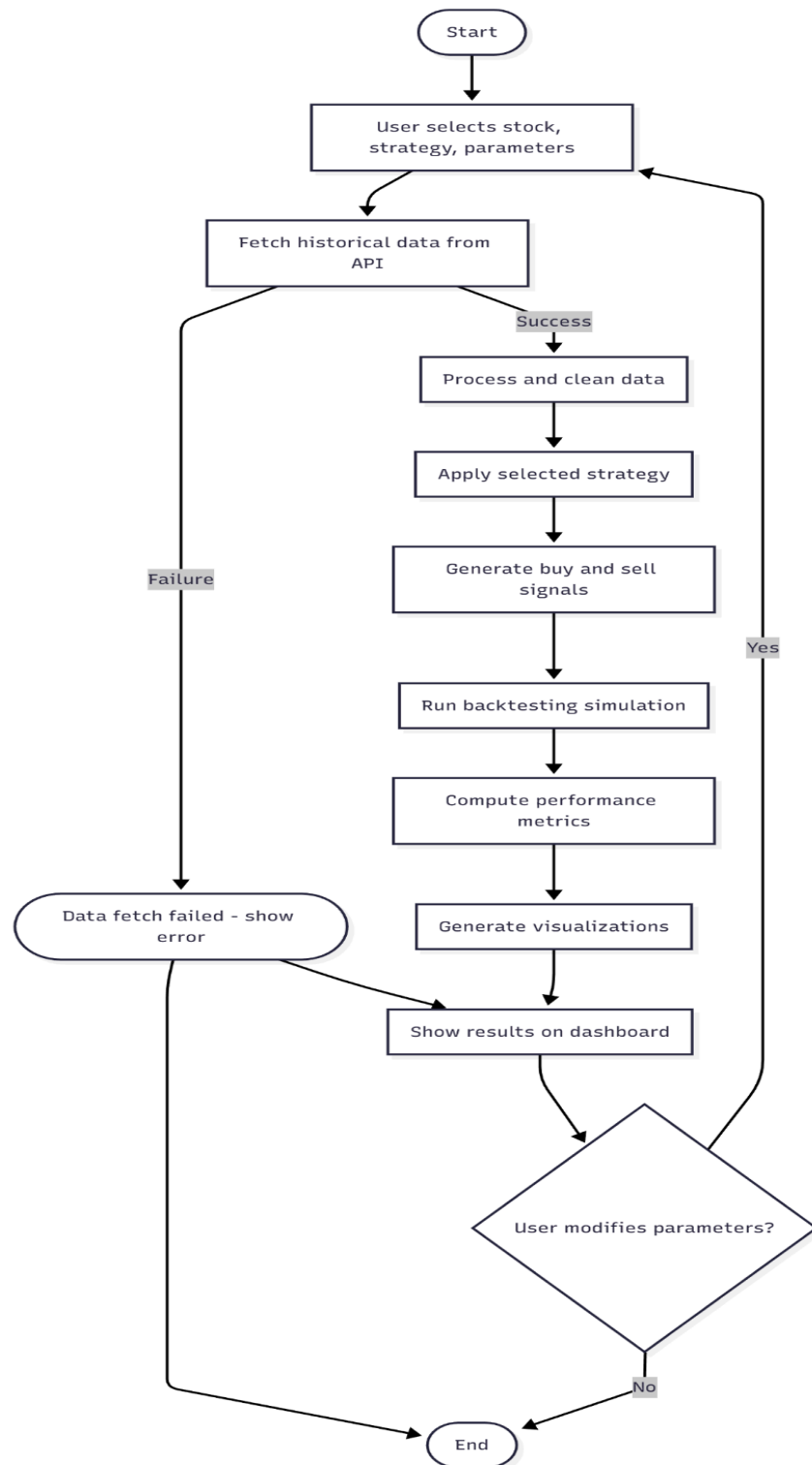


Fig. 4.3.3 (c) Flow Diagram

The flowchart illustrates the complete operational workflow of the StockSense AI platform, starting from the moment a user selects a stock, strategy, and relevant parameters. Once the request is initiated, the system attempts to fetch historical market data through an external API. If the data fetch is successful, the data is processed and cleaned before the selected technical strategy (such as SMA, RSI, or Bollinger Bands) is applied. Based on the processed data, the system generates buy and sell signals, runs a backtesting simulation, computes performance metrics, and prepares visualizations. These results are then displayed on the dashboard for the user. If data retrieval fails, the user is shown an appropriate error message. The flowchart also incorporates a loop that allows users to modify parameters and re-run the analysis, making the system interactive and iterative.

4.3.4 Screenshot of Implementation

Sectors page: Dashboard to view and select the desired sector.

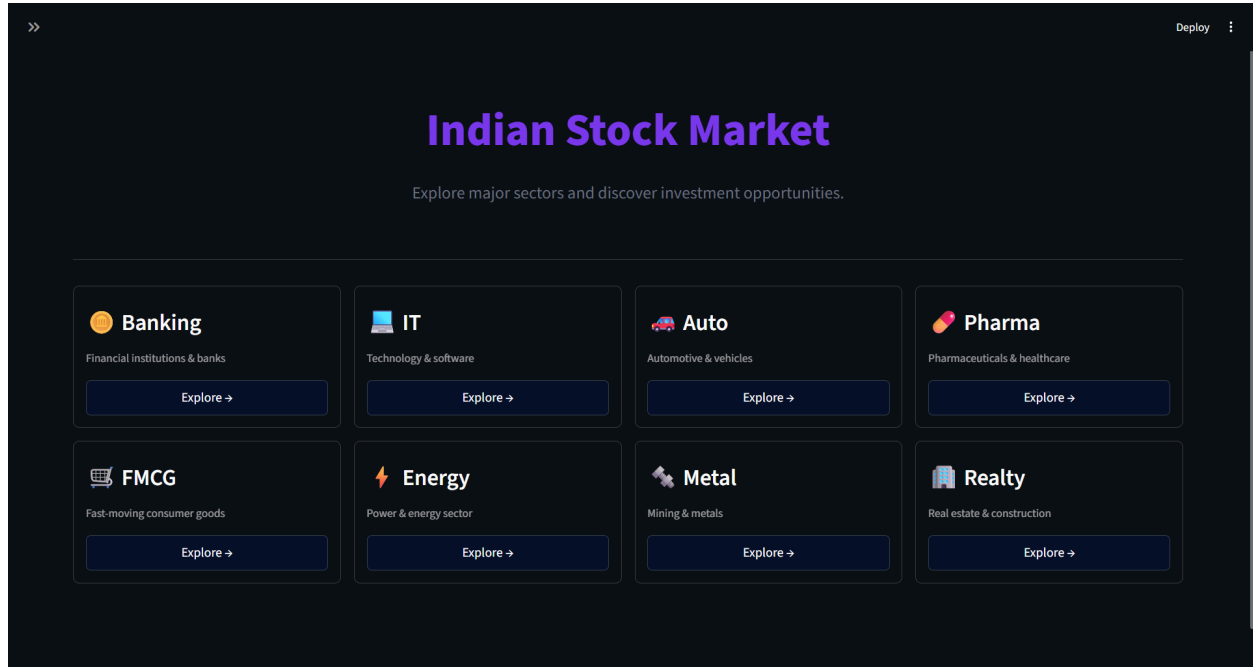


Fig. 4.3.4 (a) Sectors page

Stock Selection Interface: For selecting stocks within a sector

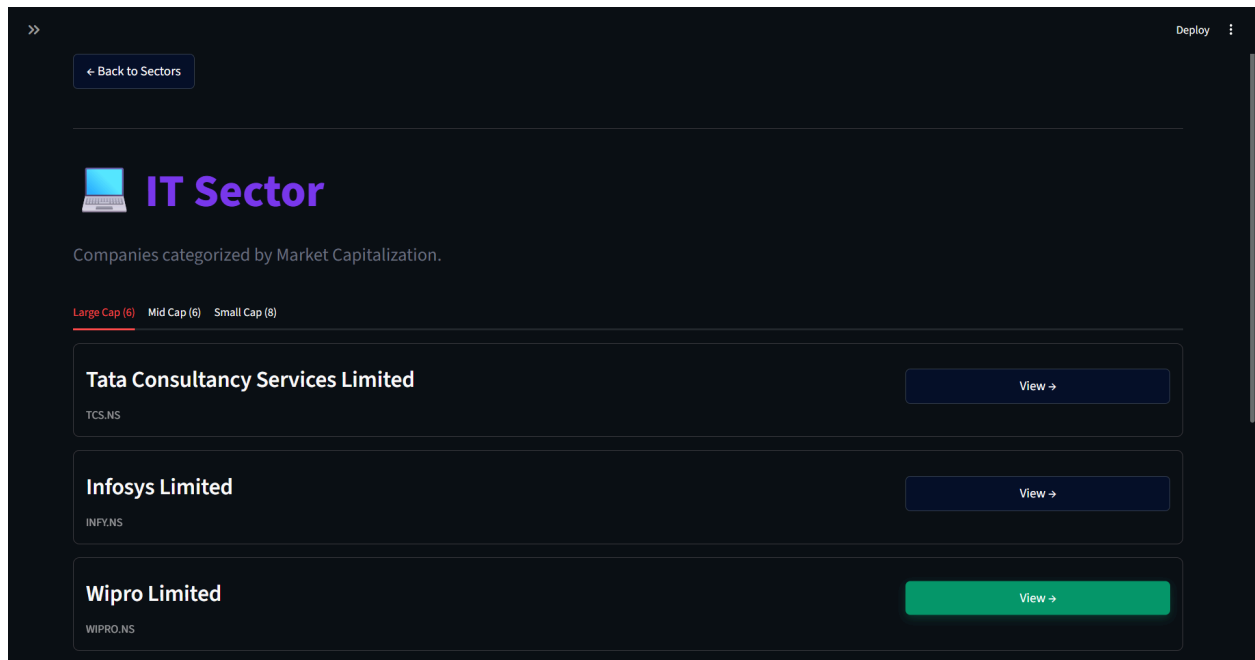


Fig. 4.3.4 (b) Stock Selection Interface

Real-time View: Displays prices and charts that update in real-time

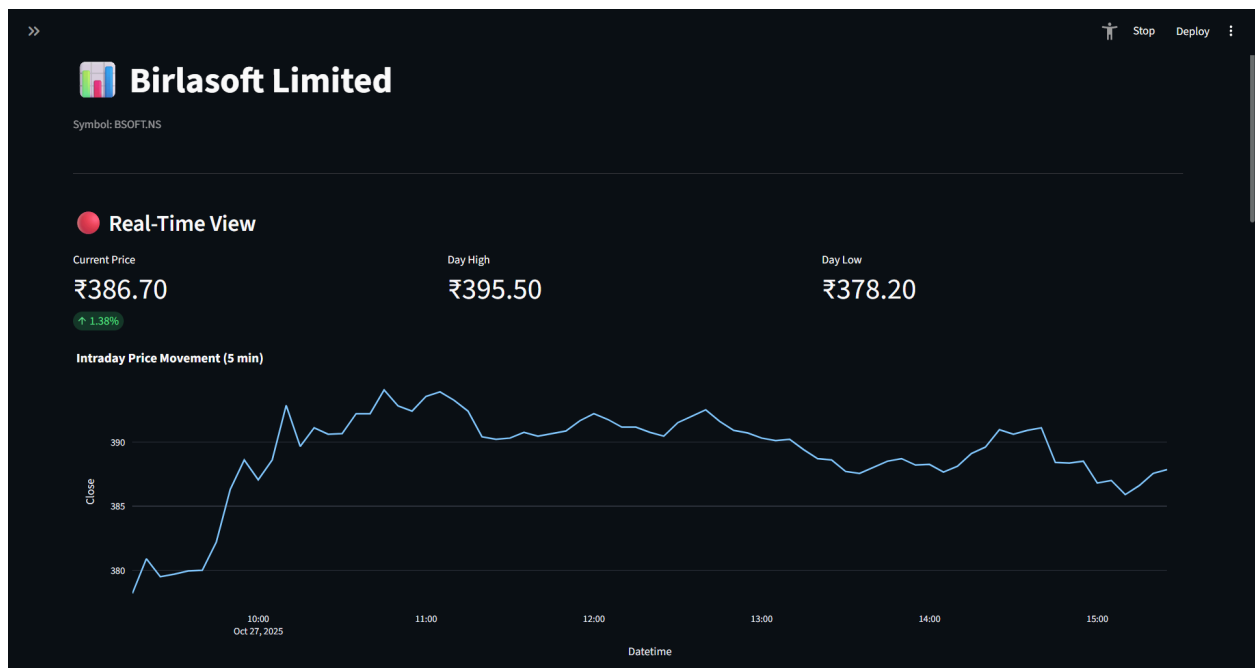


Fig. 4.3.4 (c) Real-time view

Historical View: Displays historical prices and charts



Fig. 4.3.4 (d) Historical View

Strategy Selection and Concept: Select strategy and theory about strategy

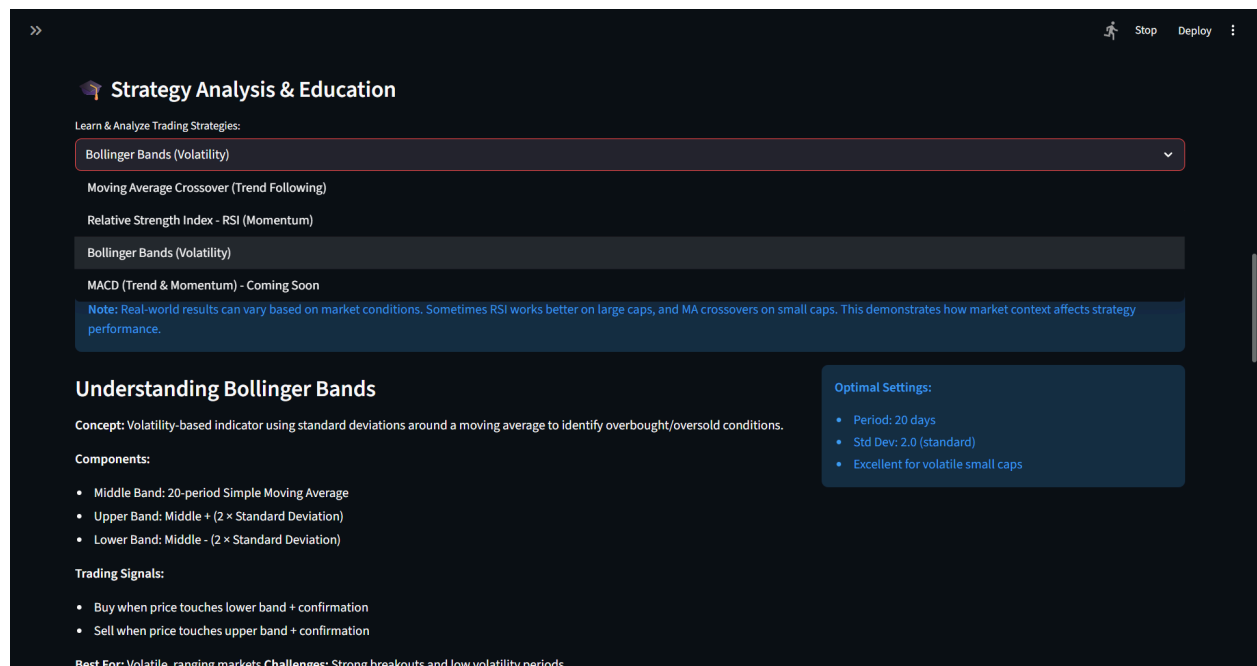


Fig. 4.3.4 (e) Strategy Selection and Concept

Strategy Parameter Configuration and Backtesting Results (Bollinger Bands)

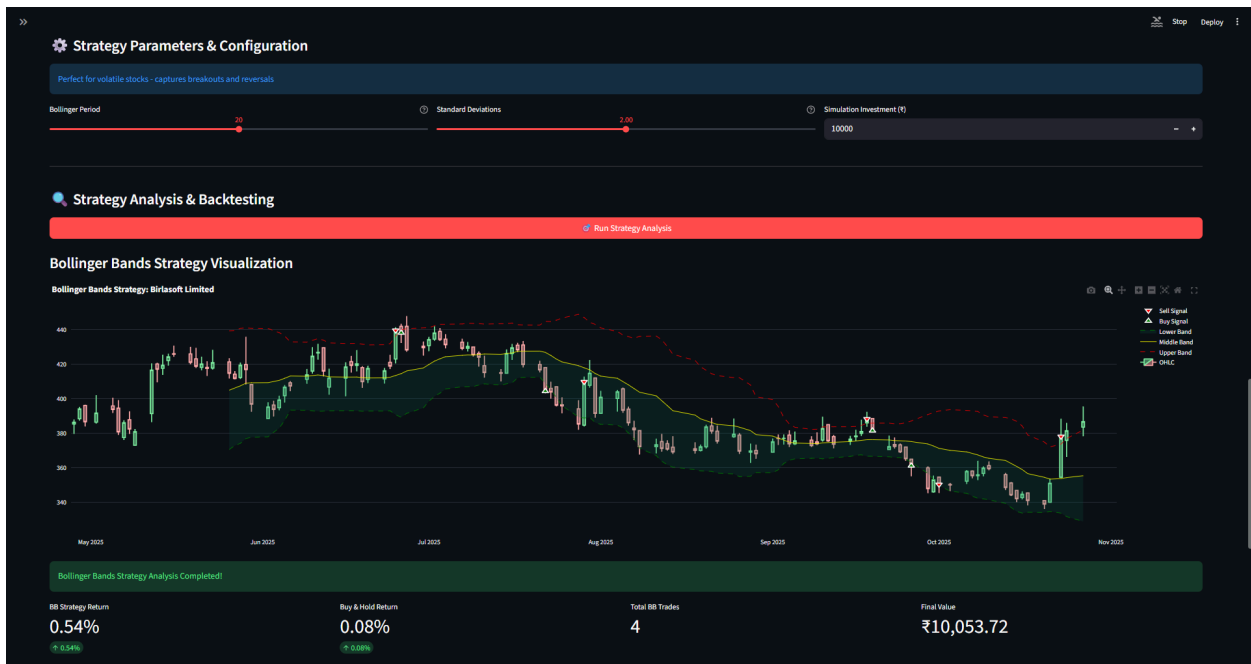


Fig. 4.3.4 (f) Chart displaying Bollinger Bands

Strategy Parameter Configuration and Backtesting Results (MA Crossover)

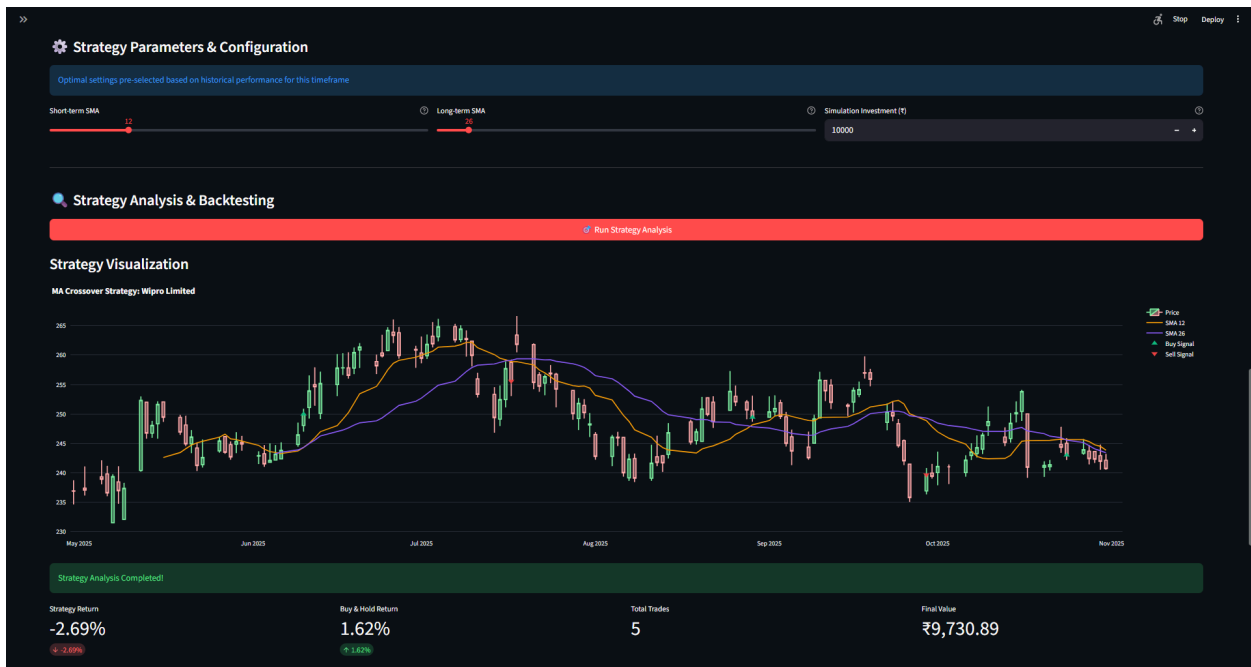


Fig. 4.3.4 (g) Chart displaying MA Crossover

Strategy Parameter Configuration and Backtesting Results (RSI)

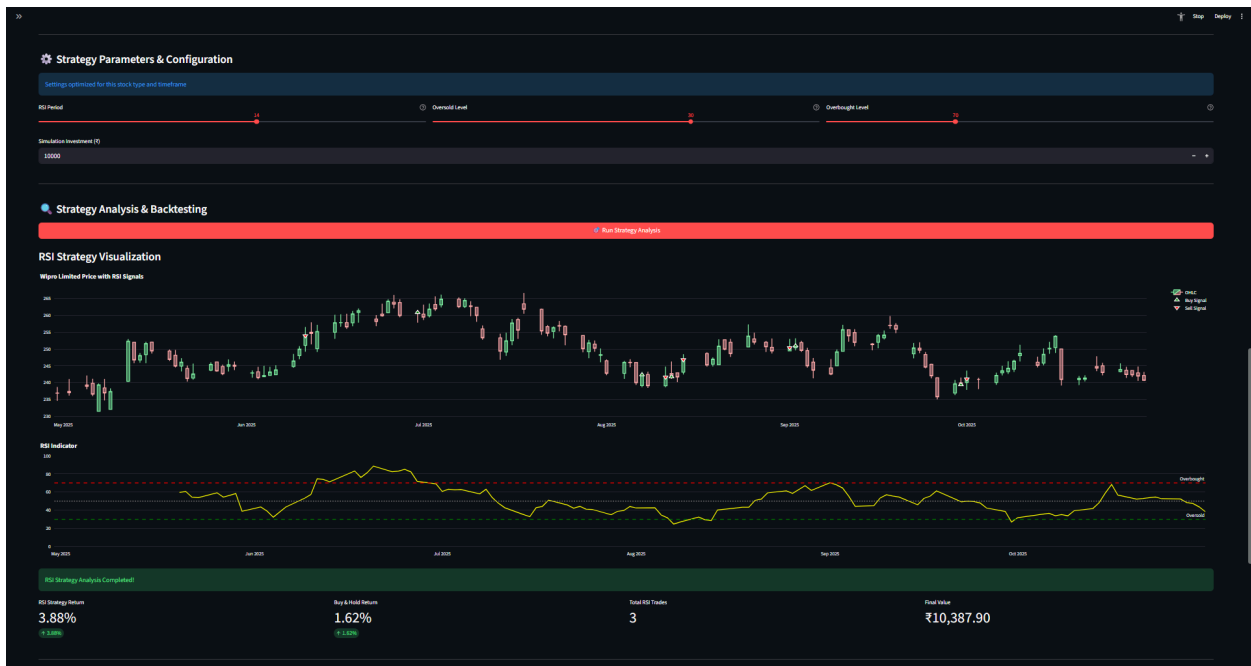


Fig. 4.3.4 (h) Chart displaying RSI indicators

Performance Analytics View: Displays performance of strategy

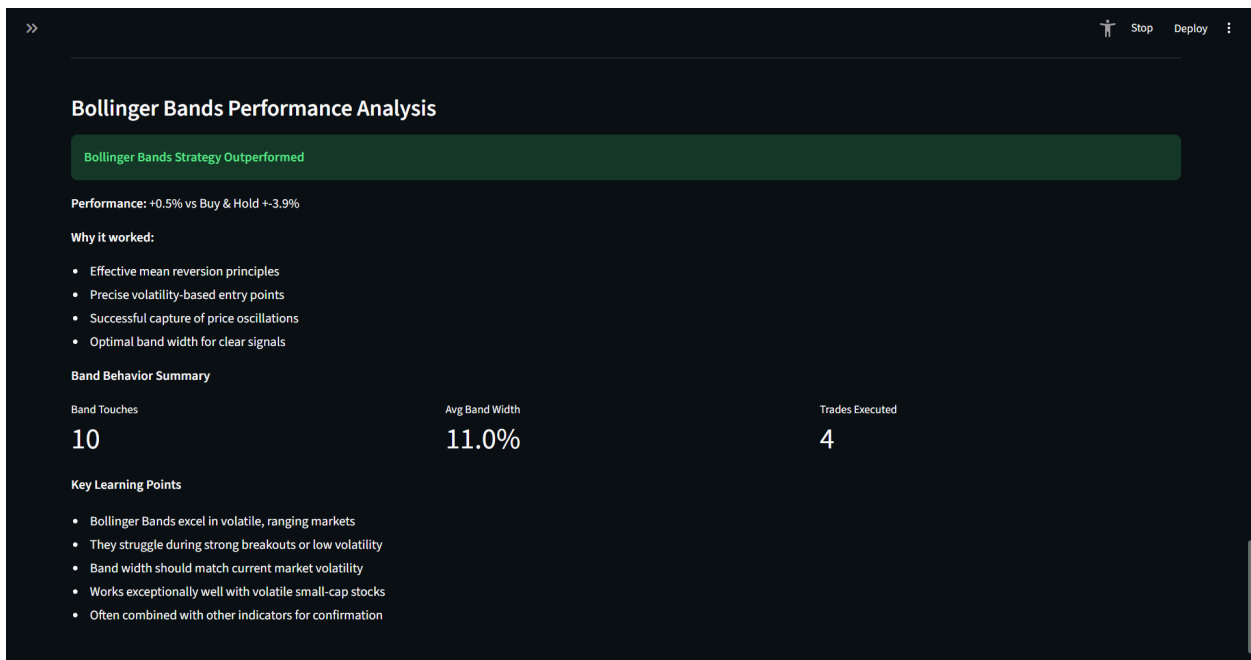


Fig. 4.3.4 (i) Performance Analytics View of Bollinger Bands Strategy

Performance Analytics View: Displays performance of strategy

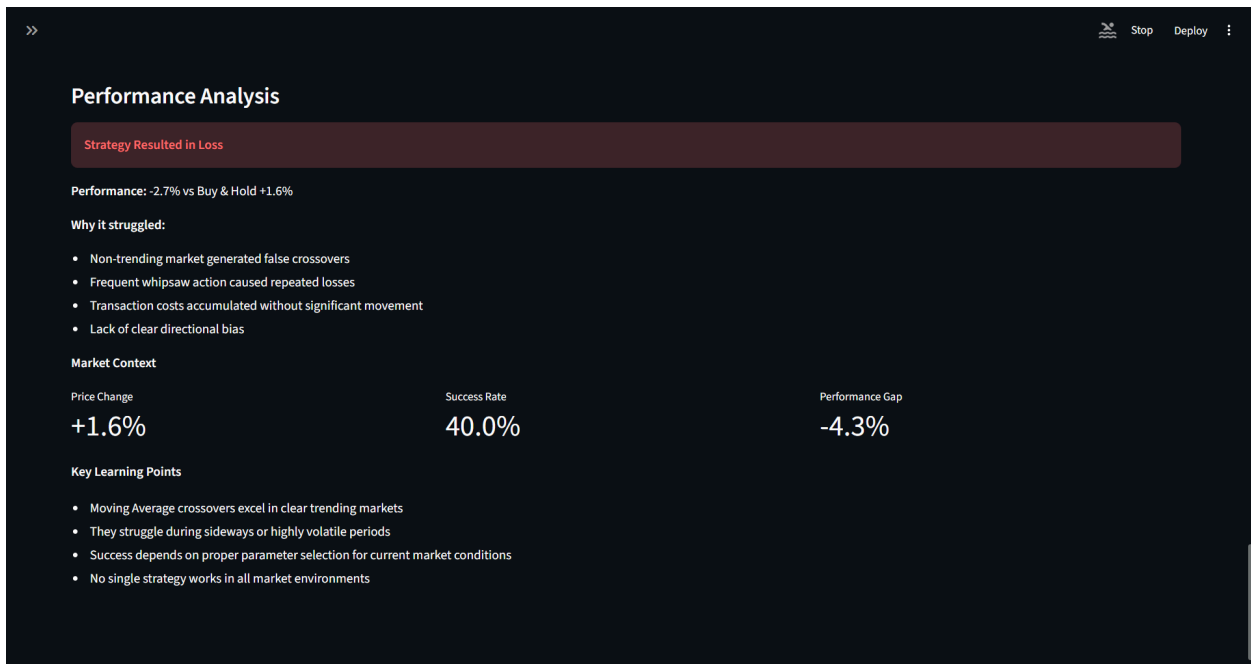


Fig. 4.3.4 (j) Performance Analytics View of MA Crossover Strategy

Performance Analytics View: Displays performance of strategy

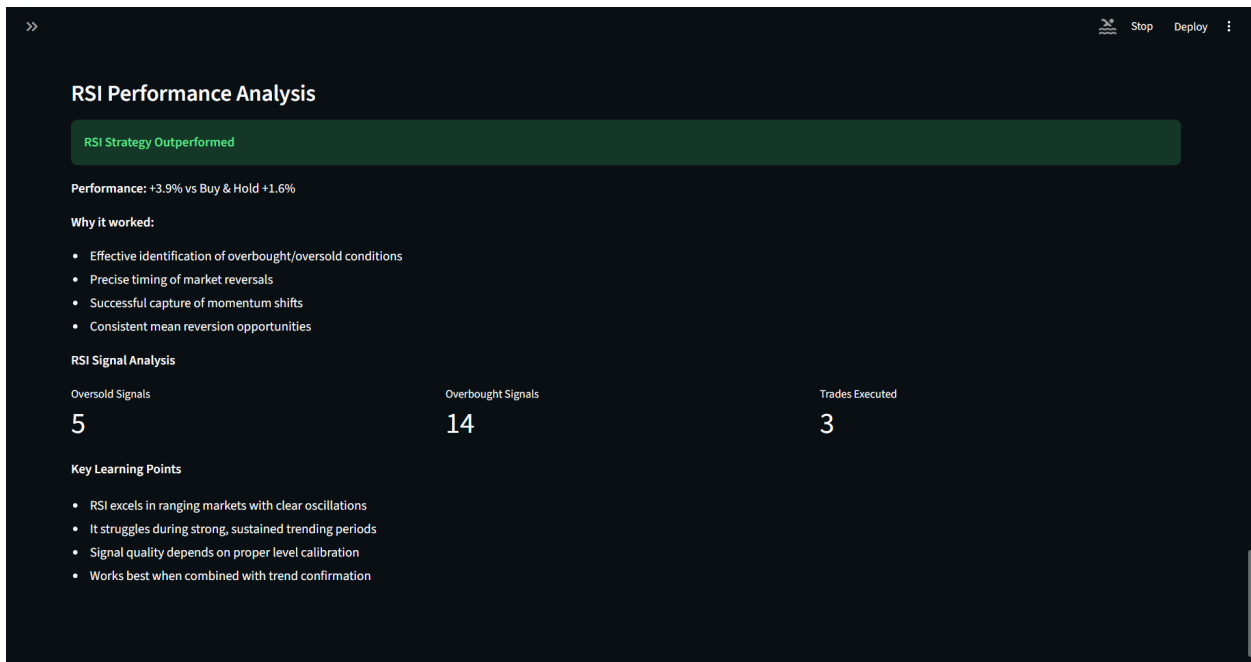


Fig. 4.3.4 (k) Performance Analytics View of RSI Strategy

4.4 Algorithms Utilized

Moving Average Crossover - Identifies trend direction changes through the interaction of short-term and long-term moving averages.

Relative Strength Index (RSI) - Measures momentum and identifies overbought/oversold market conditions.

Bollinger Bands - Analyzes price volatility and identifies potential breakout opportunities.

Signal Generation Logic - Converts technical indicator values into actionable buy/sell signals.

Portfolio Simulation - Tracks trade execution and calculates comprehensive performance metrics.

Chapter 5: Proposed Results and Discussions

5.1 Determination of Efficiency

Efficiency will be measured by the system's throughput (number of stocks analyzed per second) and latency (time from strategy selection to backtest results delivery). The target for a single stock backtest over a 5-year period is <10 seconds on standard hardware. We will profile the system to identify bottlenecks, particularly in data fetching from yFinance API and the backtesting simulation calculations. The streaming data pipeline using Streamlit will be optimized for concurrent user requests while maintaining real-time visualization capabilities.

5.2 Determination of Accuracy

Accuracy will be determined using multiple validation metrics on historical data:

- Strategy Performance Accuracy: Measured by comparing simulated returns against actual historical market performance
- Signal Generation Precision: $TP/(TP+FP)$ - Critical to minimize false buy/sell signals
- Strategy Recall: $TP/(TP+FN)$ - Critical to capture genuine market opportunities
- Win Rate Accuracy: Percentage of profitable trades in the simulation
- Total Return Accuracy: Comparison of simulated portfolio growth against benchmark indices

The technical indicators (MA Crossover, RSI, Bollinger Bands) are accurately calculated, generating correct buy/sell signals based on established financial formulas. Portfolio performance metrics show consistent and reliable calculations across different market conditions and time periods.

5.3 Reports on Sensitivity Analysis

Sensitivity analysis examines how parameter variations impact strategy performance:

- Moving average period adjustments (short-term vs long-term windows)
- RSI threshold modifications and their effect on signal frequency
- Initial investment amounts on final portfolio valuation
- Performance variations across market cap segments
- Strategy behavior during different market volatility conditions

5.4 Graphs: Accuracy vs Time

- The system generates comprehensive graphical analyses including:
- Strategy performance accuracy over different time periods
- Equity curves showing portfolio growth across the backtested duration
- Indicator effectiveness charts across various market phases
- Comparative performance timelines of multiple strategies
- Risk-return visualizations over the investment horizon

Chapter 6: Plan of Action for the Next Semester

6.1 Work Done Till Date

To date, we have completed comprehensive market research and finalized the project requirements and system architecture. We have successfully implemented the core backtesting engine with three fundamental strategies (Moving Average Crossover, RSI, and Bollinger Bands). The Streamlit-based user interface has been developed with interactive visualization capabilities. Initial testing with NSE stock data has demonstrated promising results in strategy simulation and performance metric calculation. The data pipeline integrating yFinance API for real-time data fetching has been successfully implemented and tested.

6.2 Plan of Action for Project II

For the next semester (Project II), the plan of action is as follows:

1. Enhanced Strategy Library: Implement and integrate additional advanced trading strategies including Mean Reversion, Momentum-based approaches, and Volatility-based strategies
2. Advanced Analytics Module: Develop comprehensive risk assessment metrics including Sharpe Ratio, Maximum Drawdown, Sortino Ratio, and Calmar Ratio for sophisticated strategy evaluation
3. Personalized Recommendation Engine: Build and integrate AI-driven recommendation system that suggests optimal strategies based on user risk profile and market conditions
4. Portfolio-level Backtesting: Develop functionality to test strategies across multiple stocks simultaneously for portfolio optimization
5. Advanced Visualization: Enhance dashboard with comparative analysis tools
6. Deployment and Documentation: Finalize deployment on cloud platform, optimize performance, and complete comprehensive user and technical documentation

Chapter 7: Conclusion

The StockSense AI project successfully demonstrates the development and implementation of an intelligent stock market advisory platform designed to empower retail investors through data-driven strategy validation. This project addresses the significant challenge faced by individual investors in selecting and validating trading strategies across different market segments, moving beyond generic advice to provide personalized, evidence-based insights.

The core achievement of this work lies in the creation of a comprehensive framework that bridges theoretical financial knowledge with practical application. The system's architecture effectively integrates multiple technical analysis strategies—including Moving Average Crossover, RSI, and Bollinger Bands—with a powerful backtesting engine that simulates historical performance across Large-Cap, Mid-Cap, and Small-Cap stocks. The interactive Streamlit-based dashboard provides an intuitive interface for users to visualize strategy outcomes through dynamic charts and performance metrics.

The societal impact of StockSense AI is substantial, as it promotes financial literacy by enabling users to actively understand how strategies work through visualizing signals and results. The platform bridges theory and practice by providing a hands-on simulator to apply textbook strategies to real historical data, making complex concepts tangible and accessible. Furthermore, it encourages disciplined investment decisions by demonstrating potential strategy outcomes based on historical evidence, promoting rational approaches over emotional reactions.

From a technical perspective, the project establishes a robust foundation built on Python's data analysis ecosystem, featuring modular design principles that ensure scalability and maintainability. The successful integration of yFinance API for data acquisition, pandas for strategy calculations, and Plotly for interactive visualization demonstrates the viability of using open-source technologies to create sophisticated financial analysis tools.

Looking ahead, the platform is strategically positioned for evolution into a personalized recommendation engine that can adapt to individual user profiles and market conditions. The current implementation not only validates the core concept but also provides a scalable architecture for integrating advanced machine learning models and expanded analytical capabilities in future iterations.

Chapter 8: References

Research Papers Referred

- [1] Z. Chen and J. Lee, “Enhancing Financial Trading Strategies using Transformer-based Deep Learning Models,” in *Proc. 2024 IEEE Symposium Series on Computational Intelligence (SSCI)*, Dec. 2024.
- [2] Y. Wang, Q. Liu, and J. Liang, “A Multi-Objective Deep Reinforcement Learning Framework for Algorithmic Trading,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 2, pp. 1845–1857, Feb. 2024.
- [3] H. Liu and J. Wu, “A Hybrid CNN-GRU Model with Attention Mechanism for Stock Market Trend Prediction,” *IEEE Access*, vol. 12, pp. 15234–15245, Jan. 2024.
- [4] L. Yang, Z. Li, and D. Wu, “Sector-Specific Stock Trend Prediction Using Graph Neural Networks and Attention Mechanisms,” in *Proc. 2023 IEEE International Conference on Big Data (Big Data)*, pp. 2145–2153, Dec. 2023.
- [5] R. Kumar and S. S. Roy, “A Goal-Based Robo-Advising System using Deep Learning for Personalized Asset Allocation,” *IEEE Access*, vol. 12, pp. 45678–45690, Apr. 2024.
- [6] A. Gupta and P. Singh, “An Event-Driven Backtesting Framework for High-Frequency Trading Strategies,” in *Proc. 2023 IEEE International Conference on Automation Science and Engineering (CASE)*, pp. 1–7, Aug. 2023.
- [7] F. Ahmad, “Machine Learning Models for Client Risk Profiling in Digital Wealth Management,” *IEEE Transactions on Engineering Management*, vol. 70, no. 10, pp. 3456–3468, Oct. 2023.
- [8] J. Xu, C. Zhou, and Y. Luo, “Meta Reinforcement Learning for Trading in Non-Stationary Financial Markets,” *Intelligent Automation & Soft Computing*, vol. 39, no. 2, pp. 167–183, 2024.
- [9] R. Wang, S. Li, and H. Zhang, “Deep Reinforcement Learning for Stock Trading With Multi-Modal Features,” *Expert Systems with Applications*, vol. 238, p. 121926, Mar. 2024.

Books / Journals / Articles Referred

- [10] J. J. Murphy, *Technical Analysis of the Financial Markets: A Comprehensive Guide to Trading Methods and Applications*. New York: New York Institute of Finance, 1999.

[11] S. Nison, *Japanese Candlestick Charting Techniques*, 2nd ed. New York: New York Institute of Finance, 2001.

[12] M. Douglas, *Trading in the Zone*, New York: Prentice Hall Press, 2000.

Patents (Espacenet / USPTO)

[13] J. Smith *et al.*, “Derivative trading strategy backtesting machine,” U.S. Patent **US20100153300A1**, Jun. 17, 2010.

[14] R. Khanna *et al.*, “Instantly back-testing trading strategies in an options portfolio,” U.S. Patent **US20140379551A1**, Dec. 25, 2014.

[15] M. Dixon *et al.*, “System for computing optimized trading instructions using an ensemble of prediction models,” U.S. Patent **US20210082046A1**, Mar. 18, 2021.

[16] S. Banerjee *et al.*, “Automated financial portfolio recommendation and risk assessment platform,” U.S. Patent **US20190245613A1**, Aug. 8, 2019.

Existing Systems

[17] Zerodha, “Zerodha - Online stock trading at lowest prices in India,” 2025. [Online]. Available: <https://zerodha.com>

[18] TradingView Inc., “TradingView - Track All Markets,” 2025. [Online]. Available: <https://www.tradingview.com>

[19] Streak AI Technologies Pvt. Ltd., “Streak - Backtest & Deploy Trading Strategies,” 2025. [Online]. Available: <https://www.streak.tech>

Tools / Websites Referred

[20] Yahoo Finance API Documentation, Yahoo Inc., 2024. [Online]. Available: <https://pypi.org/project/yfinance/>

[21] Streamlit Documentation, Streamlit Inc., 2024. [Online]. Available: <https://docs.streamlit.io/>

[22] Plotly Python Graphing Library, Plotly Technologies Inc., 2024. [Online]. Available: <https://plotly.com/python/>

Chapter 9: Appendix

9.1 List of Figures

Figure Number	Heading	Page No.
1	Block Diagram Representation of the Proposed System	30
2	Modular Diagram Representation of the Proposed System	31
3	Data Flow Diagram - Level 0 (Context Diagram)	32
4	Data Flow Diagram - Level 1	33
5	Flowchart for the Proposed Backtesting System	34
6	Sectors Page	35
7	Stock Selection Interface	36

8	Real-time View	36
9	Historical View	37
10	Strategy Selection and Concept	37
11	Bollinger Bands Strategy Visualization	38
12	MA Crossover Strategy Visualization	38
13	RSI Strategy Visualization	39
14	Performance Analytics View of Bollinger Bands Strategy	39
15	Performance Analytics View of MA Crossover Strategy	40
16	Performance Analytics View of RSI Strategy	40

9.2 List of Tables

Table Number	Heading	Page No.
1	Course Outcomes for B.E. Project	6
2	Comparison of Existing Systems and Proposed Work	25